

Mukul Hase

Senior Software Engineer · AI Systems & Desktop Platforms
San Francisco, CA

Email: mukulhase@gmail.com

Mobile: +1 (510) 988-6730

mukulhase.com · github.com/mukulhase

Eight years shipping production software end to end. At Figma, worked on the platform behind its AI agent and took design editing from prototype to production, spanning agent runtimes, model infrastructure, editor internals, and product UI; before that, owned the desktop client and its native macOS and Windows layers. Earlier, edge computer vision on Office Lens at Microsoft.

EXPERIENCE

- **Figma** San Francisco, CA
Sep 2021 – Current
Senior Software Engineer
 - **Figma Agent, agent platform & AI design editing (2025–Current)**: Worked on core platform behind Figma’s AI agent: Agent orchestration across Claude, Codex, and OpenCode runtimes, sandboxed agent execution, MCP tool interfaces, streaming protocols with disconnect and completion semantics, and agent prompt, configuration, and release management. Integrated a Cerebras-hosted in-house model that cut generation from minutes to 5 seconds (30x faster, now serving 80% of traffic) and built the model routing, picker, and fallback layer behind it. Improved reliability across a nondeterministic system: reduced the top failure mode from 4% to 0.5% and auto-recovered 37% of common errors through retries and self-healing agents. Took prompt-to-edit, one surface on that platform, from research prototype to production, including the on-canvas editing surface and undo/commit flow. Built the team’s debug tooling and eval infrastructure. Worked across the stack: React surfaces, TypeScript and Go services, C++ in the editor core, and Python eval tooling.
 - **First Draft, prompt-to-design generation (2024)**: Moved to the AI generation team and owned First Draft through general availability. Built the design-system intelligence behind it: matching generated output to a team’s existing component library, detecting IP conflicts, and inferring component props, plus prompt suggestions and progressive loading for slow image generation. Instrumented the analytics and dashboards used to measure generation quality.
 - **Figma Desktop, Electron (2021–2024)**: Built the desktop notification feed end to end (macOS/Windows, spanning native, Node, and web layers); tray surface, push delivery, and comment previews lifted notification click-through 25%. Integrated cross-platform native spellcheck, bringing 30+ languages to Figma and FigJam. Rebuilt desktop CI/CD from scratch: signed dogfood and per-PR build channels that caught regressions before they reached beta. Shipped the desktop side of Figma’s UI3 redesign, launched at Config, plus trackpad haptics, light mode, and the tab bar.
- **Microsoft** Redmond, WA
June 2018 – Sep 2021
Software Engineer II
 - **Whiteboard sharing feature, Microsoft Teams**: Built UI/UX in the Teams desktop app (React, GraphQL, Electron, Apollo).
 - **Edge computer vision for document cleanup, cropping & classification**: Built a parametric image rendering pipeline generating realistic synthetic document images for training and scoring, Python tooling for data collection/filtering/processing, a data-collection and training pipeline on AML, and voting/annotation/analysis tools. Ported models to ONNX/CoreML/TFLite and ran them natively on Android and iOS.
 - **Native mobile development, Office Lens (Android & iOS)**: Camera stack, Kotlin re-architecture, GPU shaders (Metal & OpenGL), custom UI animations, and a 10x improvement in document deletion/update time.
- **FreeAgent Software** Saratoga, CA
May – July 2017
SWE Intern (Bay Area startup)
 - **Full-stack web development**: React, GraphQL, Express. Built a merge feature, led an incremental-ID to UUIDv4 key migration, and built a filtering sidebar with multi-property constraints across query APIs and UI.

PROJECTS

- **WhatsAPI**: Python library for WhatsApp Web, 1500+ GitHub stars, deployed on PyPI. [github]
- **Others**: Pebble Messenger (Shake2talk over Facebook chat, 260 users); 3-Sweep (Photoshop-style 2D-to-3D image manipulation) [github]; WebGL 3D Game; IIIT’s event portal (personalized calendars by student “groups,” integrated with college LDAP).

AWARDS

- 2018 · Dean’s List, IIIT, 4th Year
- 2017 · 1st Place, NBO Hackathon
- 2016 · 2nd Place, Microsoft Code.Fun.Do
- 2016 · Dean’s List, IIIT, 2nd Year
- 2015 · Dean’s List, IIIT, 1st Year

EDUCATION

- **IIIT Hyderabad** · B.Tech Computer Science · CGPA 8.34/10 · 2014–2018

SKILLS

- **Languages**: TypeScript/JavaScript, Python, Go, C++/Objective-C++, Kotlin, Swift, Rust
- **AI & Agent Systems**: LLM agent orchestration, MCP tooling (Claude, Codex, OpenCode), sandboxed multi-agent execution, SSE streaming & chunked responses, model routing & fallback (SageMaker, Cerebras, Fireworks, Anthropic), eval harnesses & observability
- **Desktop Platform**: Electron internals, native macOS/Windows (AppKit, Obj-C++, C++), code signing & notarization (Apple, DigiCert HSM), release engineering
- **Web, Backend & Infra**: React, GraphQL, Node/Express, Protobuf, GitHub Actions, Bazel, Terraform, AWS, Datadog/Sentry